

Acoustic Droplet Vaporization Threshold: Effects of Pulse Duration and Contrast Agent

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Abstract—The use of superheated liquid perfluorocarbon droplets encased in albumin shells has been proposed as a minimally invasive alternative to current treatment of cancer by means of occlusion therapy. In response to an applied acoustic field, these droplets, which are small enough to pass through capillaries, vaporize into large gas bubbles that subsequently lodge in the vasculature. This technique, known as acoustic droplet vaporization (ADV) has been shown to successfully reduce blood flow *in vivo*, but for *in situ* conditions where attenuation is present, lower acoustic frequency and ADV threshold may be desirable. Thus, two methods to lower the ADV threshold at a lower 1.44 MHz were explored. The first part of this study investigated the role of pulse duration on ADV. The second part investigated the role of inertial cavitation (IC) external to a droplet by lowering the IC threshold in the host liquid with the presence of Definity[®] contrast agent (CA). The threshold was found to be 5.5–5.9 MPa for short microsecond pulses and decreased for millisecond pulses (3.8–4.6 MPa). When CAs were present and long millisecond pulses were used, the ADV threshold decreased to values as low as 0.41 MPa.

I. INTRODUCTION

EMBOLOTHERAPY describes the obstruction of blood flow in the vasculature, which is caused intentionally for therapeutic purposes. It is currently used in hemorrhage control and has potential utility in cancer therapy, either as an occlusive treatment to effect tissue infarction [1] or in the future as an adjunct to current treatments such as radiofrequency ablation. A possible implementation of embolotherapy was patented by Apfel [2], who suggested the use of superheated immiscible liquid droplets. Variations of such droplets have been evaluated already as potential ultrasound contrast agents [3] in which submicron sized droplets of liquid perfluorocarbon (PFC) were in part vaporized prior to injection into the body [4]. Another formulation of droplets that are larger in size (1–10 μm), also becomes superheated upon entering into the

body. When exposed to ultrasound, these droplets undergo acoustic droplet vaporization (ADV), which is the transition of droplets into gas bubbles in response to the effects of an applied acoustic field [5]. The diameters of the resulting bubbles are five to six times [5] those of the original liquid droplet and are large enough to lodge in the vasculature. ADV has been successfully shown *in vivo* to produce gas emboli and reduce blood flow [6].

Embolization by ADV is advantageous because it potentially can be performed transcatheterally with essentially no damage outside the treatment region. However, the applied ultrasound must be able to overcome attenuation from tissue when the target is deep inside the body. Low frequencies are advantageous to reduce the effects of attenuation, which should increase the range of depths and available aperture at which ADV can be accomplished. However, previous studies have shown that the pressures required to produce ADV increase with decreasing frequency [7]. Although the application is therapeutic and the bioeffects of high amplitude exposure perhaps an acceptable consequence, it remains favorable to minimize the amplitudes required to produce the desired effect for the robustness of the technique and from the patient and the logistic perspective. Hardware necessary to generate high pressures may be expensive and/or large for feasible use in the operating room.

In order to use low frequency ultrasound, other parameters may be changed to lower the ADV threshold. It may be possible that inertial cavitation (IC) is linked to ADV, in which case the opposite relationship would exist. Because IC threshold decreases with decreasing frequency, ADV threshold may also lower at a low frequency when it is associated with IC. The IC threshold in the presence of various PFC droplets was indeed lower at low frequencies when using long millisecond pulses [8]. The use of long pulses also may be advantageous because IC is a stochastic process, and the probability that IC occurs increases the longer ultrasound is applied [9].

This study investigates the effects of pulse duration and IC on ADV. The pressures required to induce each phenomenon is found for ultrasound exposures of microsecond to millisecond pulse durations at a low frequency of 1.44 MHz. Additionally, contrast agents (CAs), which have a low IC threshold, may lower the ADV threshold if in fact the two processes are linked. Therefore, a second experiment measures the ADV threshold when the droplets are in the presence of CAs.

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TABLE I
SUMMARY OF EXPERIMENTAL CONDITIONS.¹

Experimental condition	Pulse duration study		Contrast agent study	
Static	Agents:	droplets	Agents:	droplets and CA
	Detection:	B-mode	Detection:	B-mode
	Pulse seq.:	passive detector single 20 μ s–20 ms tone bursts, multiple pulses	Pulse seq.:	passive detector single 20 ms tone bursts
Flow	Agents:	droplets	Agents:	droplets and CA
	Detection:	B-mode	Detection:	B-mode
	Pulse seq:	single 20 μ s tone bursts	Pulse seq:	single 20 μ s and 20 ms tone bursts
			*Control:	CA only, 20 ms

¹Pulse duration experiments examined the ADV threshold when only droplets were present. CA experiments examined the threshold when both droplets and CA were present. A control experiment in which only CAs were exposed to 20 ms tone bursts is also listed.

II. MATERIALS AND METHODS

Experiments investigating the effects of pulse duration on ADV threshold and ADV in the presence of CA were conducted in both a static environment and in flow. Table I summarizes the experimental conditions, including acoustic parameters and method of detection for ADV or IC. Details are described subsequently in this section.

A. PFC Droplets

As described in [5], dodecafluoropentane (DDFP) (09-6182, Strem Chemicals, Inc., Newburyport, MA) was combined with a 4 mg bovine albumin (A3803, Sigma-Aldrich, St. Louis, MO)/mL saline solution. The mixture then was shaken for 30 s in a vial mixer (Vialmix, DuPont, Billerica, MA) for droplet emulsification in which the DDFP liquid droplets become encased in albumin shells. The DDFP droplet will remain as a liquid at room temperature and, although DDFP has a boiling point of 29°C, will remain a liquid even at body temperature at 37°C. At 37°C, DDFP is a superheated liquid that is stabilized against spontaneous evaporation due partially to the albumin shell [5]. All solutions of droplets were pushed through a series of syringe filters (Whatman, Maidstone, UK) so that the maximum droplet diameter was 6 μ m [6] in order to accommodate *in vivo* conditions in which droplets must be able to pass through capillaries. A droplet concentration (as determined from samples counted using a hemacytometer imaged under microscopy) of approximately 5.3×10^7 droplets/mL was used in all experiments by diluting the stock droplet solution in deionized water. The dilution then was stored in the refrigerator between experiments.

B. Contrast Agent

In the CA experiments, various concentrations of Definity[®] contrast agent (Bristol-Myers Squibb Medical Imaging, North Billerica, MA), which are microbub-

bles consisting of PFC gas in phospholipid shells, were added to solutions with the same concentration of droplet as mentioned above. The concentrations that were used were 10^3 , 10^4 , or 10^5 microspheres/mL and were determined according to appropriate dilutions of the concentration of Definity[®] specified in the package insert. Vials of Definity[®] were activated several days prior to the experiments and were reused for separate measurements. PFC gas replaced the headspace as Definity[®] was aspirated from the vial for each experiment. Unless otherwise specified, further mention of CA will refer to the use of Definity[®].

C. Acoustics

Single tone bursts of 20 μ s, 100 μ s, 500 μ s, 1 ms, 5 ms, 10 ms, and 20 ms in duration and 1000 tone bursts (20 μ s each in duration) at a pulse repetition frequency (PRF) of 500 Hz were transmitted at a carrier frequency of 1.44 MHz with an annular transducer (63 mm focal length, 63.5 mm aperture in diameter, 63.5 mm radius of curvature, Etalon 940501, Lebanon, IN). Pressure waveforms at the spatial peak of the annular transducer were calibrated with an in-house fiber-optic probe hydrophone [10]. Pressure levels ranged from ambient pressure to 6 MPa peak rarefactional pressure (P_r).

To achieve the long pulse durations used in these experiments, a “master” function generator (Hewlett Packard 33120A, Palo Alto, CA) gated a second “slave” function generator (Hewlett Packard 3314A, Palo Alto, CA). The “slave” function generator then output a 1.44 MHz sinusoid to an amplifier (ENI A300, Rochester, NY), which then was sent to the annular transducer. The “master” function generator also triggered the appropriate detector to begin recording data.

The purpose of the first set of experiments was to explore the effect of pulse duration on ADV threshold; thus, the effects of pulsing parameters described above were measured. In the second set of experiments, which explored

the influence of CAs on ADV, only single, 20 μ s and 20 ms tone bursts were used.

D. Experimental Setup for Static Experiments

The focus of the annular transducer was positioned inside a closed vessel made of thin-walled dialysis tubing (Spectra/Por®, Spectrum Laboratories Inc., Laguna Hills, CA), which was replenished with a new solution of droplets or droplets with CA between each threshold measurement. A piece of sound absorbing rubber (SOAB, BF Goodrich, Jacksonville, FL) was placed at an angle on the distal side of the tube to minimize reflection and the possibility of standing waves. Experiments were performed in 37°C deionized water (Ex 7, ThermoNESLAB, Newington, NH). A schematic of the setup is shown in Fig. 1(a).

Two methods of detection were used, although not concurrently in the actual experiment [Fig. 1(a)]. One of these methods used a diagnostic ultrasound imaging system (Diasonics VST Series, Milpitas, CA) with a 10 MHz linear array positioned approximately 30° from the annular transducer, to record B-mode images for detection of bright echoes from large bubbles created by ADV. The second method was to detect the acoustic emissions from IC, which was achieved with a passive 5 MHz single element transducer (Valpey-Fisher, Hopkinton, MA), positioned 90° from and confocal with the annular transducer. An oscilloscope (Lecroy 9310L, Chestnut Ridge, NY) was triggered with each transmitted pulse and captured the radiofrequency (RF) signals associated with any acoustic emission or scattering. The detectors were positioned as mentioned previously to minimize interference from the direct wave of the annular transducer. Similarly, because of acoustic interference from the linear array, the two detectors were used separately.

Each ADV threshold measurement was conducted as follows. A new sample was loaded into the closed vessel, leaving space at the top for gas expansion. The vessel then was connected to a holder, which oriented the vessel vertically and positioned it so that it coincided with the focus of the annular transducer. Single tone bursts or multiple tone bursts then were transmitted from the annular transducer 10–25 s apart (depending on the pulse duration) with increasing pressure amplitudes. This process was repeated five times for each acoustic parameter.

Once ADV threshold was reached, subsequent measurements at higher pressures were affected as lingering bubbles from the previous pulse facilitated further ADV. Vaporized droplets provided additional cavitation nuclei in the acoustic field of subsequent pulses if they adhered to the tube wall or if enough bubbles were produced that they could not clear the beam before the next pulse. Because we were concerned only with measuring the ADV threshold, the interference from remaining bubbles with successive pulses at super-threshold amplitudes did not affect the determined threshold.

Table I shows that both solutions (droplets and droplets with CA) were performed in a static environment. The

majority of pulse duration experiments involving only droplets were performed in a static environment, although a limited number of experiments were performed in flow for a comparison of the two environments. However, a static environment is not optimal when CAs are present because it is difficult to achieve a homogeneous distribution of CA and droplets due to a difference in buoyancy. Therefore, only a limited number of experiments with CA and droplets were performed in a static environment.

E. Experimental Setup for ADV In Flow

The circulating flow system is shown in Fig. 1(b). A solution of droplets (5.3×10^7 droplets/mL), droplets and CAs (10^3 , 10^4 , or 10^5 microspheres/mL), or CA only (10^5 microspheres/mL) were continuously stirred in a reservoir and circulated via a peristaltic pump (Masterflex® pump and speed controller, Cole-Parmer, Chicago, IL) and tubing (Tygon®, Saint-Gobain Performance Plastics Co., Akron, OH) through the focus of the annular transducer, for which a segment of dialysis tubing provided acoustic access. The pump setting was chosen to achieve a slow average flow velocity of 3.75 cm/s, which was calculated from a calibration of the volume flow rate of the pump and cross-sectional area of the tube, in order to maximize opportunity for ADV and for comparison to static experiments. Pulsatility of the flow was reduced by including an in-line capacitor in the flow system. The solution, which was first immersed and allowed to equilibrate to the 37°C tank water, was changed between each experiment. For each experiment, two pulses were transmitted 1 s apart at a specific amplitude, and the next pair of pulses were fired after 5 s. This cycle was repeated with increasing pressure amplitudes in both coarse (~ 170 kPa) and fine steps (~ 40 kPa) to encompass a large range (0.22–4.43 MPa) and to refine the threshold measurement in a small range (0.22–1.24 MPa), respectively. These experiments were repeated five times. Thus, 10 measurements were taken at each pressure in each of the two amplitude ranges, although 20 measurements existed for pressures that were equivalent in the separate ranges.

Only one detection method was used in these experiments. In flow, moving scatterers would introduce additional noise into the RF signals that are detected by the passive transducer, significantly reducing the signal to noise ratio (SNR). Therefore, B-mode images were used to make threshold measurements.

To maintain a nominally homogenous distribution of CA and droplets and to correctly characterize their behavior in concert, the majority of CA experiments were performed in flow. These solutions were exposed to 20 ms tone bursts, although an additional experiment of 10^5 microspheres/mL and droplet solutions were exposed to 20 μ s pulses for a comparison of the effect on threshold due to short-pulse duration in the presence of CAs. An experiment in which only droplets were exposed to 20 μ s tone bursts also was performed as mentioned in the previous section.

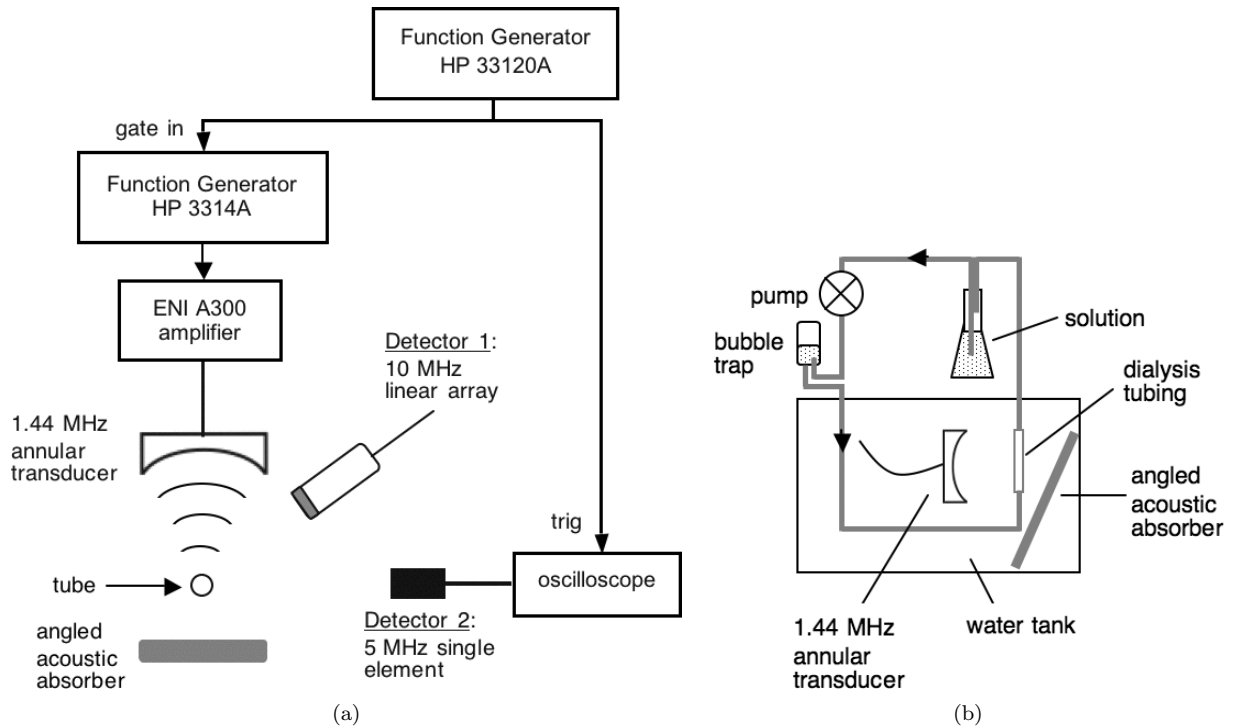


Fig. 1. (a) Top view of the experimental setup for measurements performed in a static environment. The therapeutic transducer and the associated electronics necessary to create millisecond pulses are shown. Both methods of detection also are illustrated and labeled as Detector 1 and Detector 2, though they were not used concurrently. (b) Schematic (side view) of the experimental setup used in studies measuring ADV threshold in the presence of contrast agents in a flow environment. Solutions contained either droplets, droplets and 10^3 , 10^4 , or 10^5 microspheres/mL of Definity®, or 10^5 microspheres/mL of Definity®. Experiments were performed in 37°C deionized water.

Although CAs are echogenic and provide substantial signal, scattering from bubbles with larger cross-sectional area produced by ADV are distinguishable in B-mode. The focus of the 10 MHz linear array was positioned on the proximal side of the tube and the output power was maintained at a low setting, thereby reducing the signals from CA microbubbles while still detecting signal from ADV bubbles, and reducing the possibility of destroying CA and/or affecting the threshold measurement. Because the ADV threshold is low at a frequency of 10 MHz [7], the linear array itself also is capable of vaporizing droplets.

F. Analysis of B-mode Images

B-mode images recorded during the pulse duration and CA studies were analyzed for echoes from ADV bubbles, which appear as bright pixels in the image when compared to the background. Detection for the presence of these echoes was evaluated in terms of mean echo amplitude (MEA) in a given region of interest (ROI) and is described in [5]. The ROIs, each consistent in size for a given study, were positioned next to the focus of the annular array in the image and in the direction of movement of the bubbles produced by ADV. For the static pulse duration study, the ROIs were positioned above the focus so that rising bubbles could be detected, and for the CA study where the droplets and CA were flowing, the ROIs were positioned downstream of the focus. The signal level at each pixel in the ROI then was linearized assuming a simple log com-

pression. The MEA, or the mean of the decompressed pixel signal levels in an ROI, then was calculated and compared to a baseline MEA value when no pulses were transmitted in order to detect a rise in MEA, and therefore, the occurrence of ADV. Separate explanations of how this analysis for the two studies was evaluated are described in further detail in Section III.

III. RESULTS

A. Analysis for Pulse Duration Measurements

1. *Detection with Imager:* As described previously, the MEA for each B-mode image was computed. In this pulse duration study, the MEAs for all images in a given cineloop were averaged. A baseline mean MEA then was subtracted to compute the relative echo amplitude (REA), which then was plotted as a function of peak P_r (Fig. 2). A large change in REA indicates the ADV threshold. Therefore, the derivative of the REA versus P_r curve was taken to detect a change in slope. The point at which the change in slope exceeded a certain criterion was designated the threshold. This criterion was established empirically from a small subset of the data and was subsequently applied to all data. The ADV thresholds then were plotted in terms of pulse duration (Fig. 3).

Each data point found in Fig. 2 shows the result of consecutive pulses that increase in amplitude; therefore, these

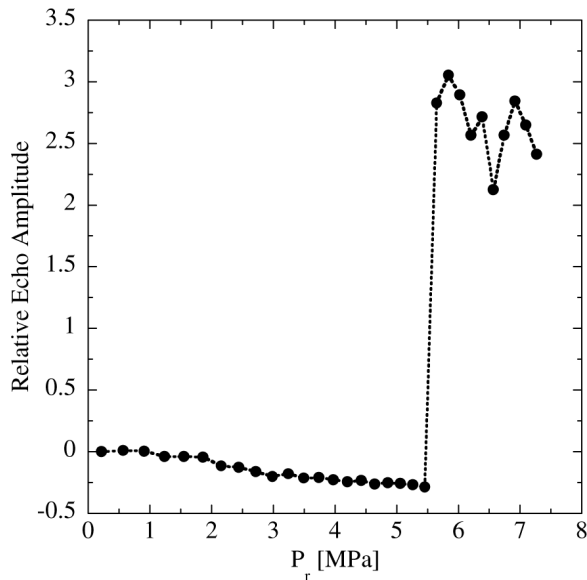


Fig. 2. Example of REA against P_r for 500 μ s pulses. The sharp change in slope at approximately 5.5 MPa reflects a rise in REA due to bubbles created by ADV.

plots reflect the REA as a function of both time and pressure amplitude. If REA before the apparent ADV event is an indication of the droplet concentration, Fig. 2 shows a decrease in droplet concentration as a function of time, which is expected as DDFP droplets have a higher density (C_5F_{12} , 1.6 kg/m³) than water and, therefore, would settle to the bottom. However, no correlation was found between any apparent variation in droplet concentration and ADV threshold found in the data.

The placement of the ROIs vertically above the focus in the B-mode images was important in order to detect rising bubbles. At near-threshold amplitudes, long millisecond pulses created echogenic scatterers, which could be misinterpreted as echogenic bubbles if they did not fall with gravity. Three frames in a B-mode cineloop illustrating this process are shown in Fig. 4(a). One hypothesis is that these sinking scatterers are DDFP droplet aggregates similar in concept to aggregated forms of CAs during an acoustic pulse [11]. Further experiments would be needed to test this hypothesis. The formation of such scatterers, henceforth referred to as “sinking scatterers”, only occurred during millisecond pulses. For super-threshold amplitudes, ADV was confirmed with the persistence of bubbles after insonification. Fig. 4(b) shows three frames after a super-threshold pulse was transmitted. Echogenic scatterers seemed to disperse above and below the focus due to radiation force (Frame 3), but scatterers in this case moved against gravity, confirming the production of ADV bubbles (Frames 10 and 20).

2. Passive Detection: A 5 MHz, single-element transducer recorded the acoustic emissions and scattering to detect IC. IC is defined as the growth and the subsequent collapse of a gas bubble during an ultrasonic wave. Its acoustic emission is characterized by the appearance of broadband noise. To specifically identify such an event,

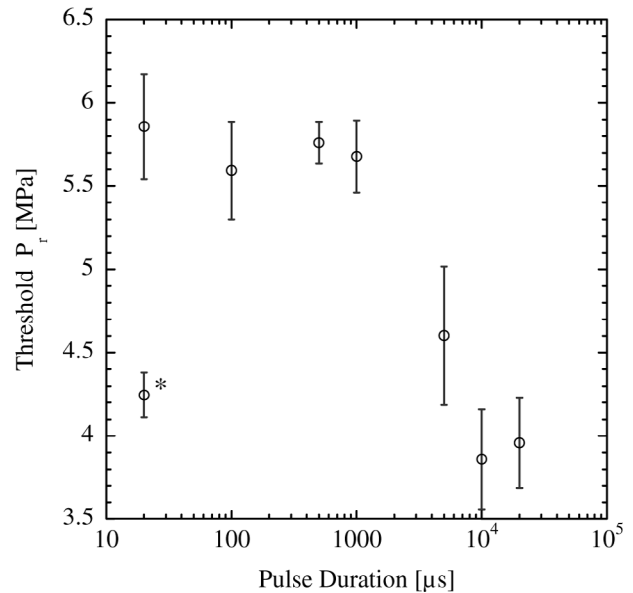


Fig. 3. Threshold in P_r as detected in B-mode images. Error bars reflect the standard error of the mean. (*) Refers to 1000 pulses (20 μ s each in duration) at a PRF of 500 Hz. All other data points are the thresholds for single pulses with pulse durations indicated along the abscissa.

the Fourier transform was computed for the entire RF waveform, and two frequency bands were analyzed by the techniques described in [12], in which the root-mean-square (RMS) was calculated to quantify spectral content. These values then were plotted as a function of P_r . To detect broadband noise, a frequency band between 2 and 2.4 MHz was chosen because it lies between the fundamental frequency at 1.44 MHz and the second harmonic at 2.88 MHz. A smaller more specific frequency band (between 0.7 and 0.74 MHz) was chosen to detect the subharmonic frequency at 0.72 MHz, which is an indicator of the presence of bubbles. Bubbles in this case may be created either by cavitation or ADV.

Two curves then were generated for each experiment. The RMS values for the two frequency bands described above were plotted as a function of P_r . Fig. 5(a) is a plot of the RMS values of the subharmonic range. Unlike the plots measuring REA (Fig. 2), the plots for spectral content varied from experiment to experiment due to variability in the number of inertial cavitation events that may occur with each tone burst and also between the different pulse durations. Therefore, a criterion computed from the baseline or prethreshold RMS value was used for each separate experiment. A histogram for each curve like that in Fig. 5(b) was used to determine the baseline value. Because the increase in acoustic emission within a band is significant, a bimodal distribution should exist. RMS values for a baseline value then are determined and used as the criteria for ADV threshold as indicated by the dotted line [Fig. 5(a)] and arrow [Fig. 5(b)].

Because of the statistical nature of IC, some events occurred at low amplitudes. For example, a spike in RMS that met the criterion for ADV may have occurred but

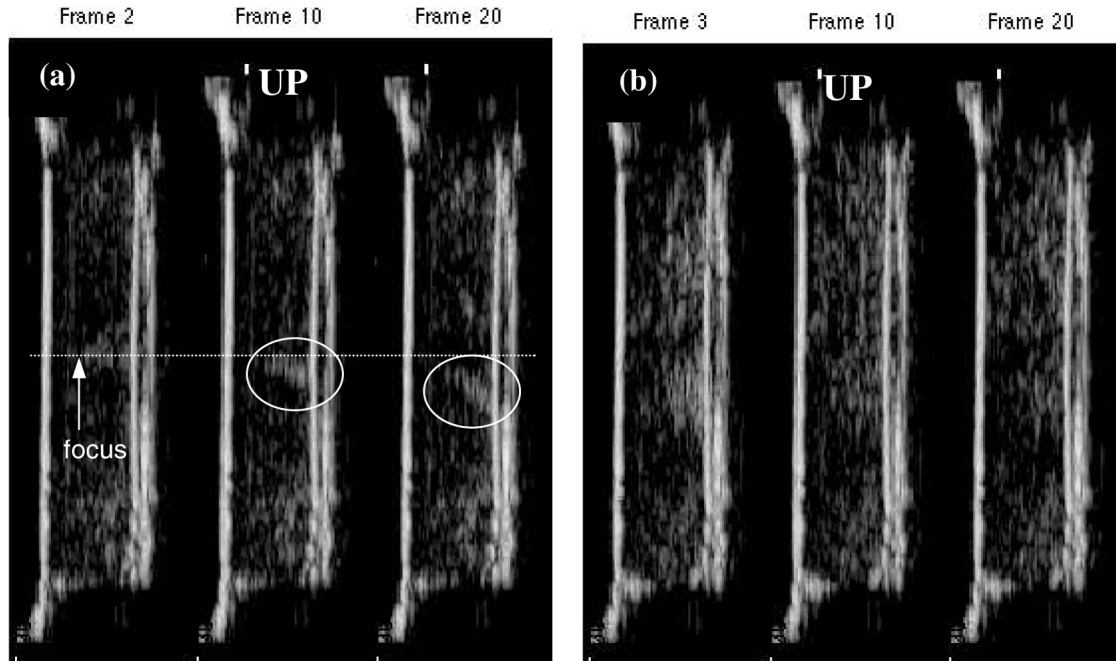


Fig. 4. (a) Sequential frames of a B-mode movie show effects of a 20 ms pulse at 1.86 MPa P_r . Frame 2 immediately follows the tone burst in Frame 1, during which acoustic interference was seen and thus is not shown here. The tube was oriented vertically, and the imager as well as the annular transducer was positioned to the left of these images. Echogenic scatterers, nonbubbles, sink toward the bottom of the tube as indicated by the circles in Frames 10 and 20. (b) Sequential frames of a B-mode cineloop show effects of a 20 ms pulse at 3.74 MPa P_r . Frame 3 immediately follows the tone burst, during which acoustic interference was seen. Acoustic interference may span from one to two frames; in this case, interference spanned from Frame 1 to 2. Due to radiation force, the echogenic scatterers were pushed against the tube wall, then dispersed above and below the beam. However, the scatterers in these images rise toward the top against gravity, indicating the creation of bubbles, and therefore, the occurrence of ADV. The reader is referred to the electronic version for the corresponding movies in which the bubbles can be better detected by their motion.

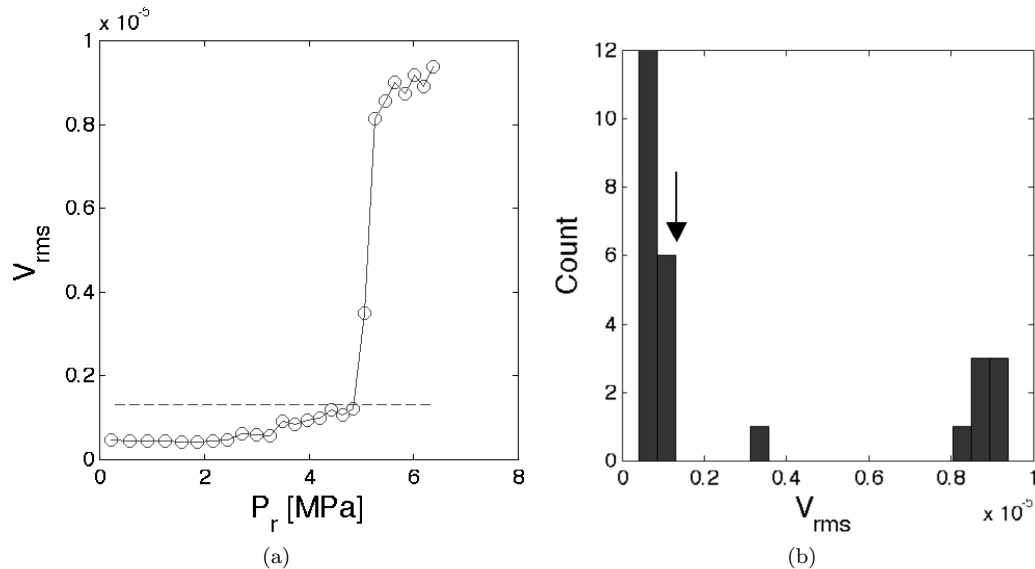


Fig. 5. ADV threshold experiment for 10 ms tone bursts. (a) Example of the spectral content of a subharmonic frequency band in terms of V_{rms} plotted against P_r . A histogram of the V_{rms} values is seen in (b), in which the lowest V_{rms} values can be separated (↓) as the baseline values. V_{rms} values greater than the baseline value (--) are determined as above threshold.

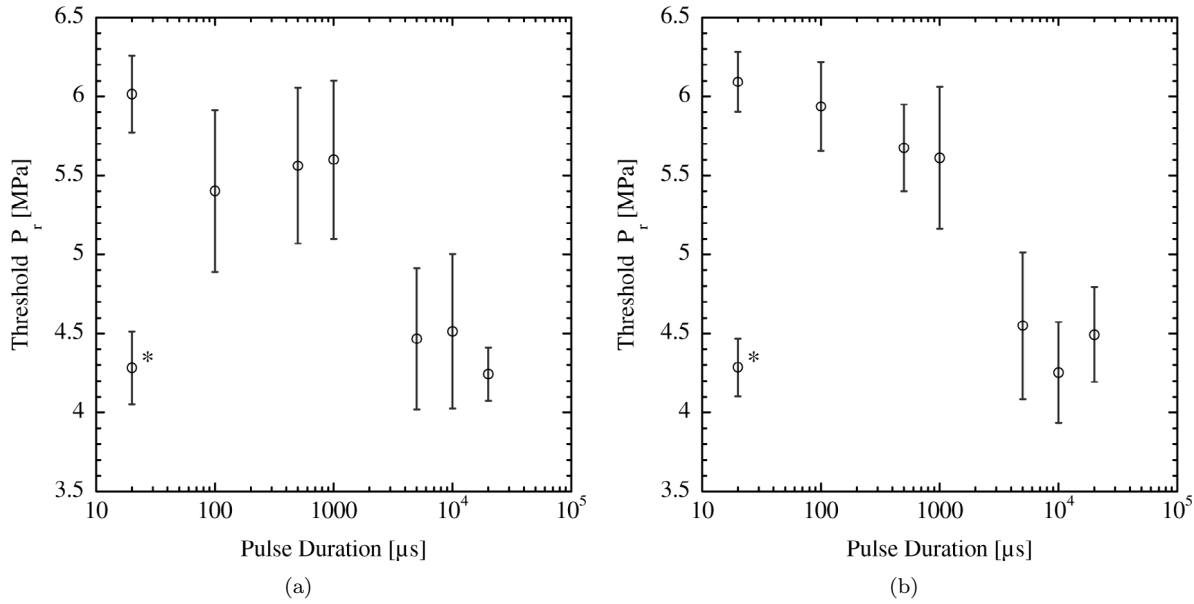


Fig. 6. Threshold in P_r as detected by the passive 5 MHz passive transducer. Error bars reflect the standard error of the mean. (*) Refers to 1000 pulses (20 μ s each in duration) at a PRF of 500 Hz. (a) Shows threshold measurements based on RMS values of the subharmonic range from 0.7–0.74 MHz. (b) Shows threshold measurements based on the presence of broadband noise.

did not recur with the following pulse at a higher amplitude. In order to choose a threshold value that would reliably predict when IC would occur, these events were not considered threshold. We required that at least two consecutive events occur to be classified as threshold. Therefore, a median filter of size three was used (1).

$$y_i = \text{median}(x_i, x_{i+1}, x_{i+2}), i = [1, N - 2]. \quad (1)$$

Once the median value y_i met the criteria, a threshold was determined. Fig. 6 shows the mean IC threshold for five measurements taken for each pulse duration. Thresholds detected by both the presence of subharmonics and broadband noise are shown.

3. *Effects of Total “On-Time” on Cavitation:* To visualize the occurrence of IC and how it relates to pulse duration and “on-time”, time-frequency plots were generated from the RF waveforms for both 20 ms pulses and for 1000 pulses (20 μ s each in duration). These two pulse sequences have the same “on-time”, which is the cumulative duration of all tone bursts. Fig. 7(a) shows three examples of scattering from 20 ms pulses at different amplitudes, and Fig. 7(b) shows four examples from the series of 20 μ s pulses, also at different amplitudes. Time-frequency plots for 20 ms pulses were generated by short-time Fourier transforms with the spectrogram function in Matlab (The MathWorks, Inc., Natick, MA) using 20 μ s segments for comparison between the two pulse sequences. RF waveforms in Fig. 7(b) were acquired in segmentation mode on the oscilloscope, and each segment was time gated to capture only the scattering during the short transmitted pulse. Fourier transforms then were taken of each segment and displayed against time in Fig. 7(b). Fig. 7(c) shows the results for 3.24, 4.20, and 4.85 MPa in Fig. 7(b) when the baseline spectrum (from the first time segment) is sub-

tracted from each of the spectra of the remaining pulses in the sequence.

B. Analysis for CA Studies

Similar to the analysis performed on the B-mode images in the pulse-length studies, ROIs were drawn in the images to detect a rise in pixel intensity, and therefore, bubble production. In flow experiments, any bubbles created by ADV would be quickly carried away after each pulse due to flow instead of buoyancy. Therefore, an ROI positioned downstream of the focus was used to capture the transit of the ADV bubbles. Fig. 8 shows the MEA of such an ROI and how it changes with time. The steady-state MEA can be seen during the first nine frames, then is followed by two transient increases in MEA that characterize the passage of bubbles, and therefore, successful ADV. Acoustic interference from the annular transducer manifesting as white bands in the B-mode image is seen in frames 10, 11, 30, and 31, and the corresponding MEAs, therefore, are omitted from the MEA versus time plots.

From these plots, a threshold can be determined. Because the occurrence of each of the two pulses is known beforehand, time intervals for a baseline MEA (first nine frames), first pulse (frames 12–15), and second pulse (frames 32–35) can be defined. Thus, MEA of the first and second pulses can be compared and evaluated for successful ADV according to a threshold criterion computed from the statistics of the baseline. Calculation of the criterion for ADV threshold is described as follows.

An accurate depiction of the baseline MEA was first obtained. Bubbles unassociated with ADV occasionally traveled through the acoustic window, were recorded in the B-mode cineloops, and were represented as aberrant spikes in the MEA versus time plots. These bubbles appeared to

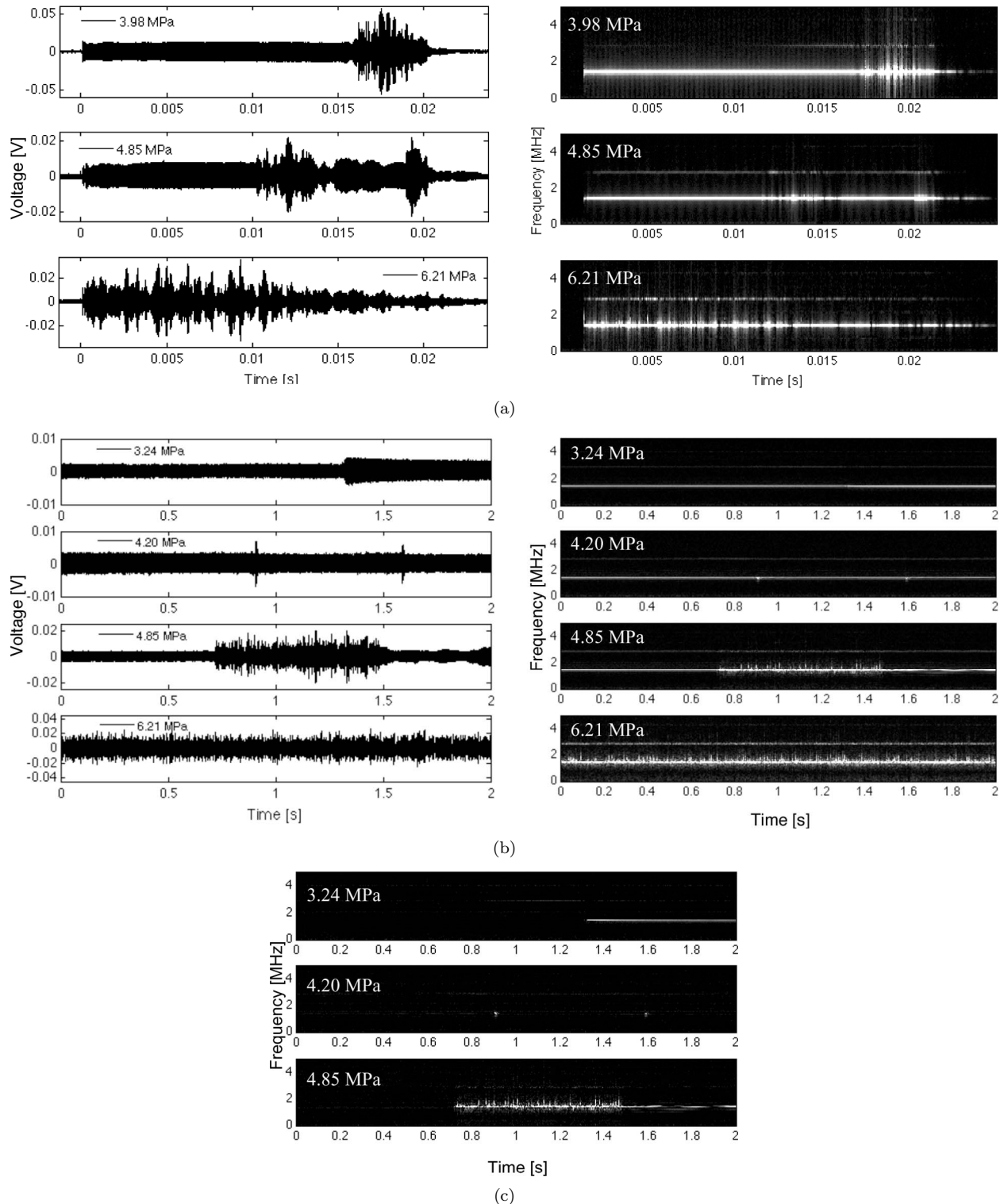


Fig. 7. RF signals are displayed for single (a) 20 ms tone bursts and for (b) 1000 pulses (20 μ s each in duration) at a PRF of 500 Hz for various pressure amplitudes. The associated time-frequency plots of the acoustic emissions are shown to the right in (a) and (b). (a) Examples of scattering at pressures above ADV threshold. (b) Four examples of RF signals, for which 3.24 and 4.20 MPa are considered below ADV threshold and 4.85 and 6.21 MPa are considered above threshold. (c) Better illustrates the change in time-frequency plots for waveforms in (b) when subtracting the spectra of the scattering from the first 20 μ s pulse. The time-frequency plot for 6.21 MPa is not shown in (c) because there is no clear baseline spectrum because IC is occurring during the entire pulse sequence.

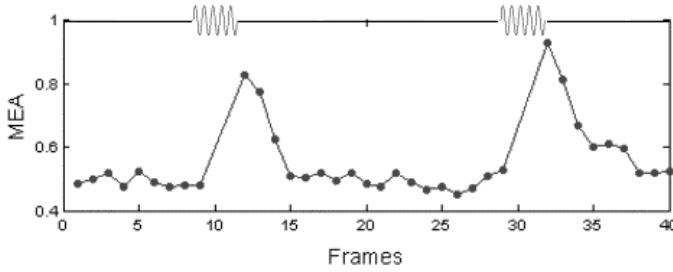


Fig. 8. Data points reflect the MEA of a given ROI through consecutive frames in a B-mode cine-loop. Statistics drawn from the mean MEA of the baseline, defined here as the first nine data points, are used as criteria for ADV detection. Pulses of the same amplitude are shown on top and illustrate the timing of the pulse firing during a given cine-loop. Frames associated with acoustic interference with the passive detector during pulse transmission are omitted (10, 11, 30, and 31).

be vapor bubbles that formed in tubing located beyond the acoustic beam. By comparing only the frames that were expected to reflect either the first pulse (frames 12–15) or second pulse (frames 32–35) to the baseline (frames 1–9), the probability that these spurious bubbles would affect ADV detection was minimized. However, these bubbles can skew the data if they appeared in the time frame used to calculate a baseline value. In order to accurately represent the baseline value, an outlier detection method was used to identify spikes in MEA in the baseline time interval and remove them from further calculations.

The box plot method, a standard method used for outlier detection, was applied to the first nine MEA data points to detect an outlier. The points were first rearranged in ascending order such that $x_1 < x_2 < \dots < x_8 < x_9$. The interquartile range (IQR) was defined as $Q_{0.75} - Q_{0.25}$, where $Q_{0.25} = \frac{1}{2} \cdot (x_2 + x_3)$ and $Q_{0.75} = \frac{1}{2} \cdot (x_7 + x_8)$. An outlier was detected if it exceeded $1.5 \cdot \text{IQR}$ above the third quartile, $Q_{0.75}$. Bubbles unassociated with ADV typically appeared as single spikes (echogenic bubbles that raise the MEA), so the criterion was applied only for the highest data point, or x_9 . Therefore, if $x_9 > 1.5 \cdot \text{IQR} + Q_{0.75}$, then x_9 was an outlier and discarded from the baseline measurement. Outliers were detected 5% of the time when using this method.

After any potential outliers were removed, the mean and standard deviation (SD) were calculated for the baseline. An ADV event occurred if the MEA associated with a pulse exceeded a criterion, which was computed from the mean and SD of the MEA baseline values and which was assessed for reliability as follows. The results ($N = 10$ or 20 as explained in Section II) according to various criteria, which differed only in the number of SDs above the mean, were compiled and plotted as a probability for each P_r and for each CA concentration (10^3 , 10^4 , or 10^5 microspheres/mL). To evaluate the reproducibility of the results, we compared the results of an applied criterion with predictions based on a binary logistic regression model (SPSS Ver.14, Chicago, IL). The predictors in the binary logistic regression were based on exogenous factors, includ-

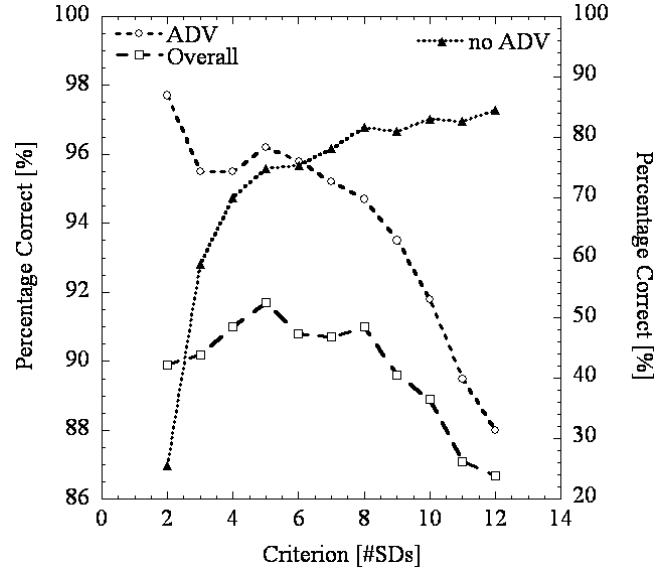


Fig. 9. Percentage of correct predictions made by a binary logistic regression model for different criteria (number of SDs). Displayed on two different scales are true negatives (no ADV) on the right ordinate and on the left ordinate, true positives as well as an overall accuracy percentage (ADV, Overall). The maximum overall percentage is 91.7% when using a criterion of five SDs above mean, meaning data, and model have the best agreement for this criterion.

ing the concentration of CA and applied pressure. Though the agreement between the results using a specified criterion was high ($> 90\%$) when the number of SD varied between three and eight, the best overall results were obtained with a criterion MEA that exceeded the mean by five SDs (Fig. 9). The model is plotted alongside the data in Fig. 10(a) using a five SD criterion and shows 50% probability of vaporization at 0.67, 0.51, and 0.41 MPa and 95% probability at 2.0, 1.92, and 1.8 MPa for CA concentrations of 10^3 , 10^4 , and 10^5 microspheres/mL, respectively. All results are statistically significant ($p < 0.0001$). A control experiment that measured the threshold in a solution with only 10^5 microspheres/mL of CA and no droplets was evaluated according to the same five SD criterion and plotted with the previous data [Fig. 10(b)]. The threshold for the control experiment is substantially higher than those for solutions containing both droplets and CA of any concentration, suggesting that the thresholds measured here are in fact due to ADV and not the CA themselves. Fig. 11(a) also further supports this argument in that a highly echogenic signal appears only with the combination of droplets and CA, whereas insonation at the same pressure amplitude actually decreases the signal level when only CA are present [Fig. 11(b)]. In addition, the thresholds measured when only CAs are present may only reflect an increase in echogenicity due to CA aggregates [11] that form in the side lobes of the acoustic beam [Fig. 11(c)].

C. ADV in Flow (Droplets Only)

A comparison also was made to observe any effects of flow on ADV. Fig. 12 shows the results of an experiment with only droplets in flow along with the ADV threshold

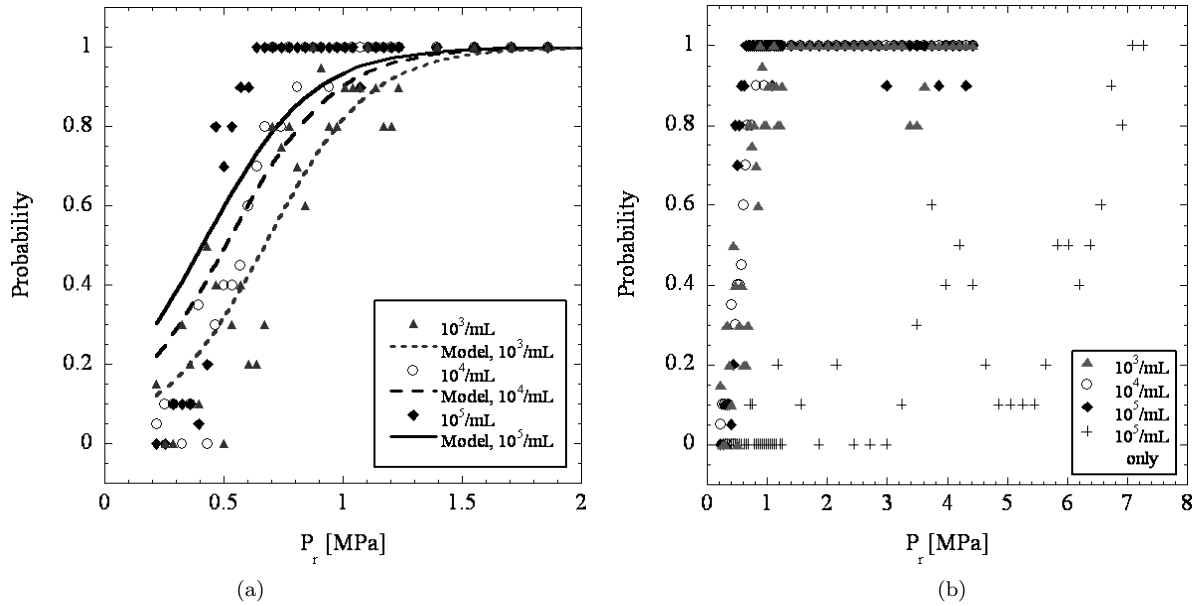


Fig. 10. (a) Probability for ADV computed in the presence of 10^3 , 10^4 , and 10^5 microspheres/mL of Definity® according to a criterion of five SD above mean. Data at a given pressure amplitude show the result for $N = 10$ or 20 measurements. The entire dataset, including results for $P_r > 2$ MPa, is not shown in order to better display the transition of probability from 0 to 1. Curve fits are taken from the regression model described in the text. (b) The entire dataset for the experiments in (a) is displayed here along with data that were collected with only CA present (+), which show a higher threshold than the solutions containing droplets. Data for (+) reflect $N = 10$ measurements.

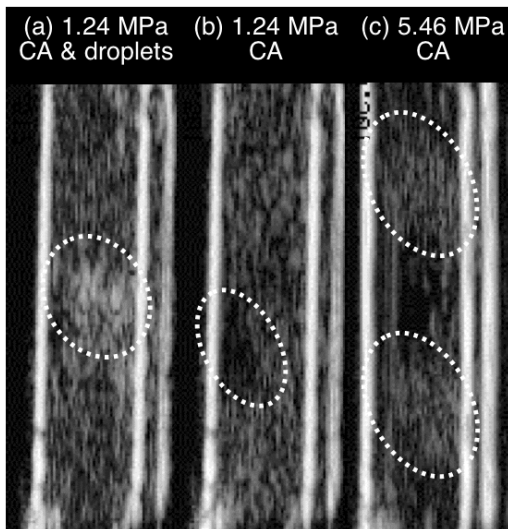


Fig. 11. Frames immediately following acoustic interference are shown when solutions containing both (a) droplets and CA and (b), (c) CA only are exposed to 20 ms pulses. ADV occurs in (a) at 1.24 MPa as indicated by the circle. At the same pressure, CAs are destroyed as indicated by low echogenicity when droplets are absent (b). Increasing the pressure to 5.46 MPa further causes high echoes in areas surrounding the focal zone, which may be the aggregation of CA microspheres in the side lobes.

found in the static experiments for 20 μ s pulses. Because of the possibility of forming echogenic scatterers other than ADV bubbles during long 20 ms pulses, it would be difficult to differentiate between the bright echoes due to bubbles and those due to sinking scatterers in flow as both move in the direction of flow. Therefore, 20 μ s pulses were used for this comparison. Fig. 12 shows that an average

flow velocity of 3.75 cm/s does not significantly affect ADV threshold for 20 μ s pulses, the shortest tone burst used in this paper. Because the annular transducer has a 6 dB lateral beam width of 1.14 mm and a single droplet will have moved 750 nm in 3.75 cm/s flow, nominally all droplets in the beam will have been exposed to the entire 20 μ s pulse. Thirty-four percent of the droplets in the 6 dB lateral beam width will have been exposed to an entire 20 ms pulse, which may result in an elevated threshold measurement due to limited detector sensitivity. Additionally, the results for an ADV threshold experiment for 20 μ s pulses in the presence of CA were plotted in Fig. 12, illustrating a reduction in threshold for short pulses as well.

D. ADV in a Static Environment (Droplets and CA)

A limited number of experiments with CAs (10^5 microspheres/mL) and droplets were performed in a static environment with 20 ms pulses because, as mentioned previously, it is difficult to differentiate bubbles from sinking scatterers in a flow environment. From these static experiments, we can confirm from B-mode cineloops that at 0.43 MPa, echogenic scatterers rise after insonification. This is the same pressure that generates echogenic scatterers in a flow experiment and is approximately 20% that of the pressure required to generate sinking scatterers in a static environment. The results from the static case support the claim that the echogenic scatterers detected in the flow environment are in fact buoyant bubbles produced by amplitudes as low as 0.43 MPa (Mechanical Index or $MI = 0.4$).

Passive detection of acoustic scattering also was used in these limited static experiments in which droplets

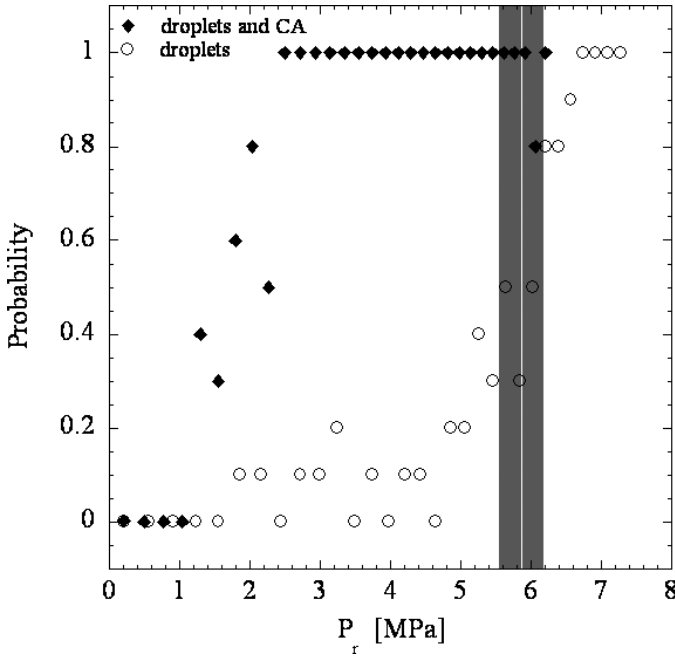


Fig. 12. Probability results for ADV using 20 μ s tone bursts. CA concentration is 10^5 microspheres/mL. Circles and diamonds are data collected from ADV in flow conditions. The white vertical line and the gray region indicate the mean ADV threshold in P_r with no CA present and the associated standard error, respectively, found in the static experiments using detection by B-mode cine-loops.

were mixed with CA at a concentration of 10^5 microspheres/mL. The apparent threshold for broadband noise in these experiments was approximately 0.8 MPa, which is just above the range of the transition of probability from 0 to 1 in the flow experiments. This difference may be due to the inhomogeneous distribution of CAs and droplets, which results from a difference in buoyancy and consequently a separation of material. The elevated threshold is not likely due to a difference in detector sensitivity because there is good agreement between the thresholds detected by the imager and passive detector in the static experiments (Figs. 3 and 6). Nevertheless, these results indicate that IC is occurring at these low pressure amplitudes.

IV. DISCUSSION

The results show that the ADV threshold can be lowered by both lengthening pulse duration as well as by adding CA. In evaluating these two methods, the occurrence and role of IC also are examined.

A. ADV Threshold with Droplets Only

From the static experiments, it appears that the ADV threshold remains relatively constant (5.5–5.9 MPa) for pulse durations shorter than 1 ms for a 1.44 MHz carrier frequency (Fig. 3). The threshold drops for pulse durations longer than 1 ms, whereupon it appears to reach a plateau (3.8–4.6 MPa). Although 20 ms may vaporize droplets at

a lower P_r than microsecond pulse durations, it may not be optimal for *in vivo* conditions because of heat deposition on overlying tissue layers. Thus, we investigated if the total “on time”, or the cumulative duration of all tone bursts, had an effect on the ADV threshold by transmitting 1000 pulses at 20 μ s duration each (20 ms “on time”) at a PRF of 500 Hz (1% duty factor). Fig. 3 illustrates that the threshold for this case (mean of 4.25 MPa) is not statistically significant from the threshold found for a 20 ms tone burst (mean of 3.96 MPa).

B. The Role of Inertial Cavitation in ADV

To investigate the role of inertial cavitation, for which broadband noise signifies the occurrence of an event, the acoustic scattering from the exposure of droplets to ultrasound was recorded and compared to the data captured by the imager. The thresholds as determined by the two different detection methods as well as the threshold trends seen in terms of pulse duration (Figs. 3 and 6) agree. A constant threshold for pulse durations less than 1 ms is seen in Figs. 3 and 6 and is consistent with the literature [13], in which the time-averaged cavitation activity in blood remained constant for pulse durations from 20 μ s–1 ms at 1 MHz. Both curves also show that the threshold drops for pulse durations longer than 1 ms. There is no indication in the data presented here nor in the literature that suggests the threshold would drop further for pulse durations longer than 20 ms [8]. This consistency among the data suggests at least that IC plays a role in ADV at frequencies below several megahertz, although it is not clear whether the inertial cavitation occurs inside or outside of the droplet in the experiments when only droplets are exposed to ultrasound.

The elevated threshold for broadband noise of 4.5 MPa ($MI = 3.8$) for 20 ms pulses as compared to the results found in the literature [8] may be explained when considering that IC and ADV are linked. With a lower f/number transducer (1 versus an f/number of 2 in [8]), fewer higher harmonics accumulate because of reduced time for high-amplitude ultrasound propagation. Because high frequencies are associated with a reduced ADV threshold, it may be more difficult for a low f/number transducer to vaporize droplets. As this difference in threshold is shown in *in vitro* experiments, transducer geometry and the effects of nonlinear propagation must be further investigated for *in vivo* applications, for which tissue provides substantial attenuation. Although the nonlinearity coefficient β is similar for both water and tissue (3.6 and 4–6.2, respectively [14]) the Goldberg number is much higher in water than in tissue [15]. High harmonics generated during propagation through water would be attenuated to some degree if instead the medium were tissue.

C. IC and “On-Time”

The decrease in ADV threshold with increasing pulse duration along with the comparable thresholds of equiv-

alent “on-time” pulses is reminiscent of the behavior of IC. It was shown that hemolysis, which reflects the quantity of IC, increased with longer pulse duration and total “on-time” [9]. It may take as few as one available nucleus to induce IC [9], and the opportunity for this to occur increases with longer pulse durations.

Examples of RF waveforms and the associated time-frequency plots are shown in Fig. 7. Given adequate time for insonification, the chances of a cavitation nucleus passing through the acoustic beam increases. At 3.98 MPa, IC occurs only a portion of the time during the 20 ms pulse [Fig. 7(a)]. The duration of IC increases with increasing amplitude and is eventually occurring during the entire pulse at 6.21 MPa. This increase in IC duration also can be seen in the multiple 20 μ s pulses at 4.85 and 6.21 MPa [Fig. 7(b)]. These amplitudes are considered above threshold, at which there is a spread in the spectral content indicating broadband noise. Fig. 7(b) also shows RF waveforms at amplitudes below ADV threshold (3.24, 4.20 MPa) that contain an interesting characteristic not seen in single 20 ms pulses. For the case at 3.24 MPa, the scattering amplitude increases at approximately two-thirds into the duration of the pulse sequence and may appear to reflect the accumulation of sinking scatterers, which was seen in B-mode cine-loops. This is consistent with an increase in scattering only at the fundamental frequency of 1.44 MHz. At 4.20 MPa, spikes in the scattering signal occur at two time points and appear to reflect the generation of only a few bubbles per spike. Although there is a slight spread into frequencies lower than the carrier frequency, it was not sufficient to meet threshold criterion for subharmonics. The change in acoustic emissions during insonification is better realized in Fig. 7(c). Because the imaging array and passive, single-element transducer could not be used concurrently, there is no direct evidence coupling the aforementioned events though they can be seen during separate experiments at the same amplitude.

D. Inertial Cavitation External to a Droplet

The possibility that cavitation outside of the DDFP droplet may trigger ADV was investigated by the addition of CA to the solution. CAs such as Alunex[®] (Molecular Biosystems, Inc., San Diego, CA) [13], [16]–[18] and Levovist[®] (Schering AG, Berlin, Germany) [17] have been shown to lower the IC threshold of the host medium and, in the case of Optison[®] (Mallinckrodt Medical, St. Louis, MO), lower it even further than the broadband emission threshold for PFC droplets [8]. At diagnostic power levels, ultrasound exposure of Definity[®] has been shown to cause bioeffects thought to be associated with cavitating microbubbles. These bioeffects include an increase in troponin T at $MI = 1.2$ [19] and petechial hemorrhages at an apparent threshold of 0.37 MPa for 1.7 MHz, or $MI = 0.3$ [20]. These thresholds are significantly lower than the cavitation threshold found with droplets present at $MI = 3.8$ for 20 ms pulses. If cavitation is linked to ADV as an initiator of the phenomenon, then there could be a reduction in ADV threshold with the presence of CAs. We see that

in Figs. 10 and 12 there is indeed a reduction in threshold with the introduction of CA. In fact, it is lowered by an order of magnitude for 20 ms pulses for the three CA concentrations investigated here and by a factor of three for 20 μ s pulses and 10^5 microspheres/mL of CA. Although IC in the host medium may not always be the trigger for ADV, these results indicate that it can be.

With a higher concentration of CA, there is a higher probability of cavitation occurring with each pulse because there are more nuclei present [21]. As such, 10^5 microspheres/mL provide more nuclei, and therefore, there is a higher chance of IC, and a higher chance that the event occurs in proximity to a droplet. The initial threshold is considered here as the amplitude in which there is at least one event [Fig. 10(a)]. Although this threshold is similar for all concentrations of CA, the transition to achieving a probability of 1 rises more quickly with increasing P_r for 10^5 microspheres/mL. The logistic binary regression model showed that the results of all three CA concentrations were statistically significant relative to each other ($p < 0.0001$), with the probability of an occurrence of an ADV event increasing with the concentration of CA Fig. 10(a). A similar result was seen with Levovist[®] and Alunex[®] CAs, each of which had an IC threshold (detected by H_2O_2 production) of 0.41 MPa independent of the concentration of CA added [17]. There was no statistical difference in the amount of cavitation activity when comparing activity increase from various dilutions of one or the other CA that were added to denucleated phosphate buffered solution. However, when Alunex[®] was added to a solution with higher gas content, a notable rise in cavitation activity resulted. The solutions in the experiments described in this paper were not degassed, and perhaps the combination of higher gas content, droplets, and Definity[®] provided substantially more nuclei to show a significant difference between the three CA concentrations.

Although the results of the CA studies in this section have been discussed in comparison to thresholds of other types of CA, they do not reflect the effects of all types of CA. It has been shown that CA oscillations differ with the type of stabilizing shell [22], [23], and we can reason from bubble dynamics that the type of gas also will affect the contrast agent behavior. However, it also has been shown in [20] that the bioeffects (petechial hemorrhages) resulting from IC occur at the same pressure amplitude for Optison[®], Definity[®], and Imagent[®] (Alliance Pharmaceutical Corp., San Diego, CA), and that the degree of bioeffects depends on the number of microbubbles and is independent of the type of CA. Because our passive detection method was used to identify the occurrence of IC, and the thresholds for IC and ADV were similar, the effect of CA on the ADV threshold may be the same regardless of the type of CA.

E. ADV Threshold In Vivo

It is noted here that the findings of these studies may not represent droplet behavior in blood. It has been shown *in vitro* that the ADV threshold increases when blood is

the host medium [5], although further investigation is required to determine the effects of blood flow *in vivo*. Additionally, the solutions in these experiments were not degassed and were expected to be either at equilibrium with air as they were circulated in the flow system, or slightly supersaturated in the static system, in which gas dissolved into the solution during refrigeration before being placed into the 37°C water bath. However, there did not appear to be a significant difference in ADV threshold between the two systems (Fig. 12). However, human arterial blood has a P_{N_2} of 571 mmHg, PO_2 of 102 mmHg and a PCO_2 of 40 mmHg [24]. Arterial blood, which is the target location for embolotherapy for cancer treatment, therefore, would have a slightly different gas content than that found in these experiments. Still, with the addition of CA and consequently an even lower IC threshold, the ADV threshold also is likely to be reduced in blood.

Additionally, for successful therapy *in vivo*, the bubbles resulting from ADV must expand to an appropriate volume to be able to occlude arterial feeder vessels to the cancer. Although there has been success in a lepus kidney model [6] as mentioned previously, verification of sufficient bubble sizes is necessary for applicability in humans.

V. CONCLUSIONS

In order to achieve ADV at a low ultrasonic frequency and reduced input energy, ADV threshold was measured as a function of pulse duration for single tone bursts and for multiple pulses fired at a given PRF. By increasing pulse duration at 1.44 MHz, both the ADV and IC thresholds decreased in a similar trend. It also was found that repetitive pulsing could induce the same effect at similar amplitudes as a single pulse if the two have equivalent “on-times”. This may suggest a link between the two phenomena, although proof of a causal relationship and determination of whether IC occurs inside or outside of the droplet remains for future investigation.

In testing for the influence of IC on ADV, further experiments were conducted to evaluate the role of IC external to a droplet by the addition of CAs. The presence of CAs greatly reduced the ADV threshold by an order of magnitude for 20 ms pulses and by a factor of three for 20 μ s pulses. Perhaps, with the use of CAs and by choice of an appropriate pulse duration that reduces ADV threshold while minimizing input power, routine ADV can be feasible at a range of useful locations in the body.

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